FUNCTION POINT CALCULATION

Project: Anora - Mental Health Journaling Application

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Timeline: 3 Months

Technology Stack: Flutter (Dart) for frontend/mobile app, Hive for local encrypted database, MentalBERT/DistilBERT for AI inference, AES-256/RSA for encryption, Federated Learning for distributed model training, Blind Mailman protocol for backend encrypted storage.

Aim

To calculate function point for Anora Mental Health Journaling Application with AI-powered sentiment analysis, encryption, and federated learning capabilities.

Theory

Function Point Analysis (FPA) is a standardized method used to estimate the size, complexity, and functionality of software projects.

Unadjusted Function Points (UFP): Calculate UFP by multiplying the number of each type of component by its corresponding weight and summing them up.

Value Adjustment Factor (VAF): A factor that adjusts the UFP based on general system characteristics (GSCs), which include considerations like performance, usability, and complexity.

$$FP = UFP * (0.65 + 0.01 * Sum\ of\ GSCs)$$

a) Number of External Inputs (EIs)

User Onboarding:

- Username: 1
- Password: 1
- Email: 1
- Profile Setup (Age, Gender, Mental Health Goals): 3

Journal Entry:

- Text Entry: 1

- Mood Rating: 1
- Tags/Categories: 1
- Date/Time: 1

Settings & Preferences:

- Privacy Settings: 1
- Notification Preferences: 1
- Theme Selection: 1

Clinician Sharing:

- Clinician Email: 1
- Share Trigger: 1

Crisis Management:

- Panic Button Trigger: 1

Federated Learning:

- Opt-in/Opt-out Toggle: 1

Total: 17

b) Number of External Outputs (EOs)

AI-Generated Outputs:

- Risk Alert Notification: 1
- Sentiment Score: 1
- Daily Mood Summary Card: 1

Clinical Outputs:

- Clinical Summary JSON (Encrypted Locked Box): 1

Export & Reports:

- PDF Report Export: 1

Error Messages:

- Invalid Credentials: 1
- Encryption Failed: 1
- Sync Failed: 1
- Invalid Input Format: 1

Total: 9

c) Number of External Inquiries (EQs)

Dashboard Views:

- View Current Mood Trends: 1
- View Weekly Summary: 1
- View Monthly Analytics: 1

Search & Retrieve:

- Search Past Journal Entries: 1
- Filter by Date Range: 1
- Filter by Mood/Tag: 1

Total: 6

d) Number of Internal Logical Files (ILFs)

User Data:

- User Profile Database: 1

Journal Data:

- Encrypted Journal Store (Hive): 1

Security:

- Key Store (AES-256/RSA Keys): 1

Analytics:

- Cached Analytics Data: 1

Total: 4

e) Number of External Interface Files (EIFs)

AI Model:

- Global Federated Learning Model (Downloaded from Server): 1

Security Infrastructure:

- Clinician Public Key Directory: 1

Backend Services:

- Blind Mailman Storage (Encrypted Blob Storage): 1

Total: 3

Information Weighting Factor

Domain	Count	Simple	Average	Complex	Calculation	Total
External Inputs (EIs)	17	3	4	6	17 × 4	68
External Outputs (EOs)	9	4	5	7	9 × 5	45
External Inquiries (EQs)	6	3	4	6	6 × 4	24
Internal Logical Files (ILFs)	4	7	10	15	4 × 15	60
External Interface Files (EIFs)	3	5	7	10	3 × 7	21
Count Total						218

Note: ILFs are assigned Complex weight (15) due to AES-256 encryption complexity.

The Fi (i=1 to 14) are Value Adjustment Factors (VAF) based on responses to the following questions:

1. Does the system require reliable backup and recovery?

Ans: Reliable Backup and Recovery

Score: 5

Our system requires reliable backup and recovery to ensure journal entries and mental health data are never lost. Patient mental health data is critical and irreplaceable.

2. Are specialized data communications required to transfer information to or from the application?

Specialized Data Communications

Score: 4

The system needs to communicate with backend servers for federated learning gradient updates and encrypted clinical report transmission via the Blind Mailman protocol.

3. Are there distributed processing functions?

Distributed Processing

Score: 5

The system implements Federated Learning where AI model training is distributed across multiple user devices while maintaining privacy. This is a core architectural feature.

4. Is performance critical?

Performance Critical

Score: 5

Real-time AI inference for sentiment analysis must run on mobile devices with limited CPU. Performance is critical for user experience in a mental health context where delays could impact usability during crisis moments.

5. Will the system run in an existing, heavily utilized operational environment?

Existing, Heavily Utilized Environment

Score: 3

The system runs on standard mobile devices (Android/iOS) with typical background app constraints and battery limitations.

6. Does the system require online data entry?

Online Data Entry

Score: 5

100% of journal data is entered interactively by users through the mobile interface. All user inputs are captured in real-time.

7. Does the online data entry require the input transaction to be built over multiple screens or operations?

Multiple Screens for Data Entry

Score: 2

Journal entry is designed for simplicity (single screen), but onboarding and clinician sharing involve multiple step workflows.

8. Are the ILFs updated online?

Online Update of ILFs (Internal Logical Files)

Score: 5

The Hive database is updated immediately upon every journal entry, mood rating, and settings change. Real-time updates are essential.

9. Are the inputs, outputs, files, or inquiries complex?

Complex Inputs, Outputs, Files, or Inquiries

Score: 4

The system handles complex encrypted data structures, JSON clinical summaries, and multi-dimensional sentiment analysis outputs.

10. Is the internal processing complex?

Complex Internal Processing

Score: 5

The system performs MentalBERT/DistilBERT inference, AES-256 encryption, RSA public-key cryptography, federated learning gradient masking, and multi-factor risk scoring algorithms.

11. Is the code designed to be reusable?

Reusable Code

Score: 4

Flutter's widget-based architecture and clean architecture pattern enable high code reusability. The local-first design allows components to be reused across platforms.

12. Are conversion and installation included in the design?

Conversion and Installation

Score: 1

Standard mobile app installation via App Store/Play Store. No complex data migration from legacy systems.

13. Is the system designed for multiple installations in different organizations?

Multiple Installations in Different Organizations

Score: 2

The system is designed for individual users but supports multiple clinician integrations. Not organization-specific but clinician-agnostic.

14. Is the application designed to facilitate change and ease of use by the user?

Facilitate Change and Ease of Use

Score: 5

The system is designed with mental health accessibility in mind, featuring Flutter's hot reload for rapid updates, modular architecture for feature additions, and an intuitive UX critical for users experiencing mental health challenges.

Value Adjustment Factor Summary

Characteristics	Score
Reliable Backup and Recovery	5
Data Communication	4
Distributed Functions	5
Performance	5
Heavily Used Configuration	3
Online Data Entry	5
Transaction Rate	2
Online Update	5
Complex Inputs/Outputs	4
Complex Processing	5
Reusability	4
Installation Ease	1
Multiple Sites	2
Facilitates Change	5
Total (Sum of fi)	55

Function Point Calculation

FP = Count-total × [0.65 + 0.01 × Sum(fi)]

FP = 218 × [0.65 + 0.01 × 55]

FP = 218 × [0.65 + 0.55]

FP = 218 × 1.20

FP = 261.6

~ 262 FPs

Lines of Code Estimation

LOC = FPs × Language Factor

For Flutter (Dart) with modern framework and packages:

Language Factor = 50 LOC per FP

(Reduced from standard 60 due to Flutter's high-level abstractions, widget reusability, and extensive package ecosystem)

LOC = 50 × 262

LOC = 13,100 Lines of Code

Conclusion

The Function Point Analysis for the Anora Mental Health Journaling Application has been successfully completed.